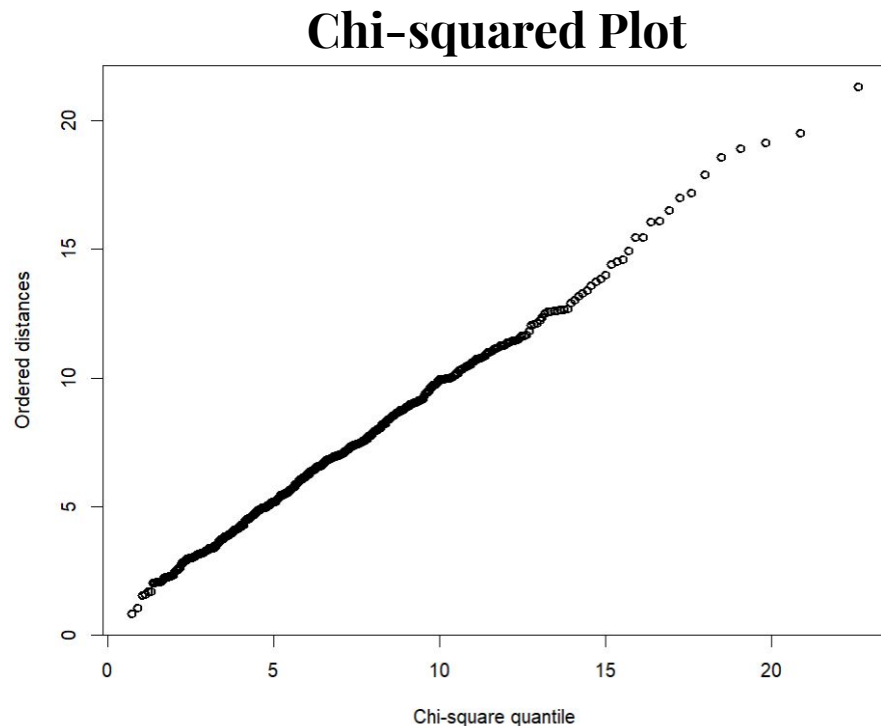


Mental Health & Social Media Balance: A Multivariate Study

Zoe Horn, Maggie LePorin, Theodore Villalva, Emmy Sitnik

Dataset and Variables

- Kaggle dataset
- Lifestyle and mental wellbeing
- Discrete, categorical, and continuous variable types
- Social media platforms
- Range of data
- Multivariate normality of our data

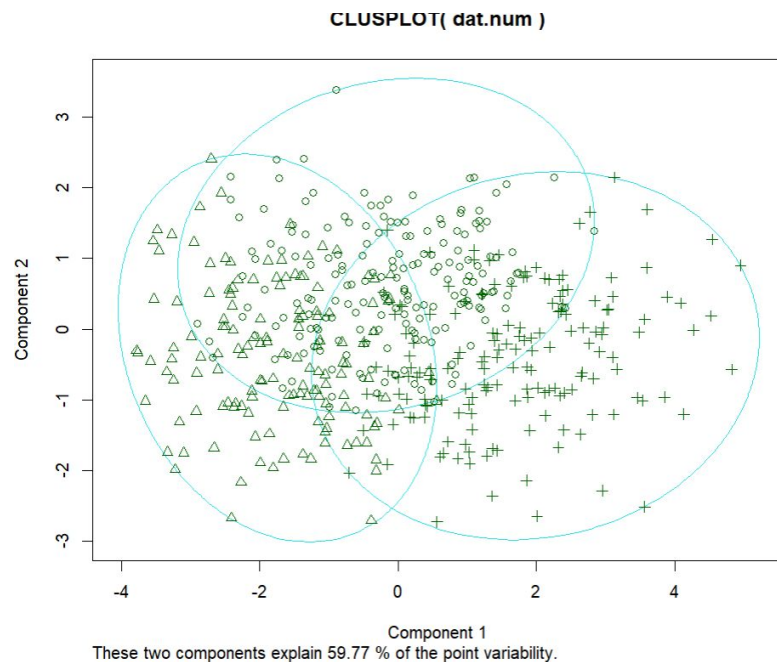
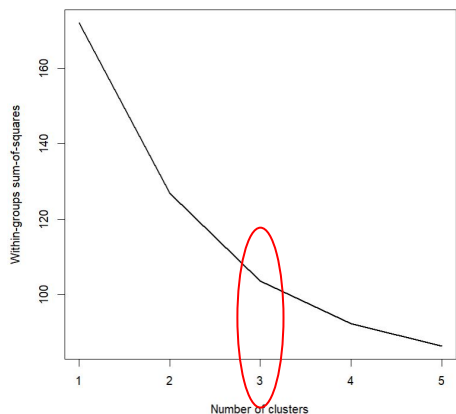


Methodologies

- Chi-squared plot for multivariate normality assumption
- Correlation coefficients
- Scatterplot matrix to visualize relationships
- Factor analysis
- Principal Component Analysis
- 3D plots for multivariate correlations
- Compare factor scores to most popular social media app

What Methods Didn't Work?

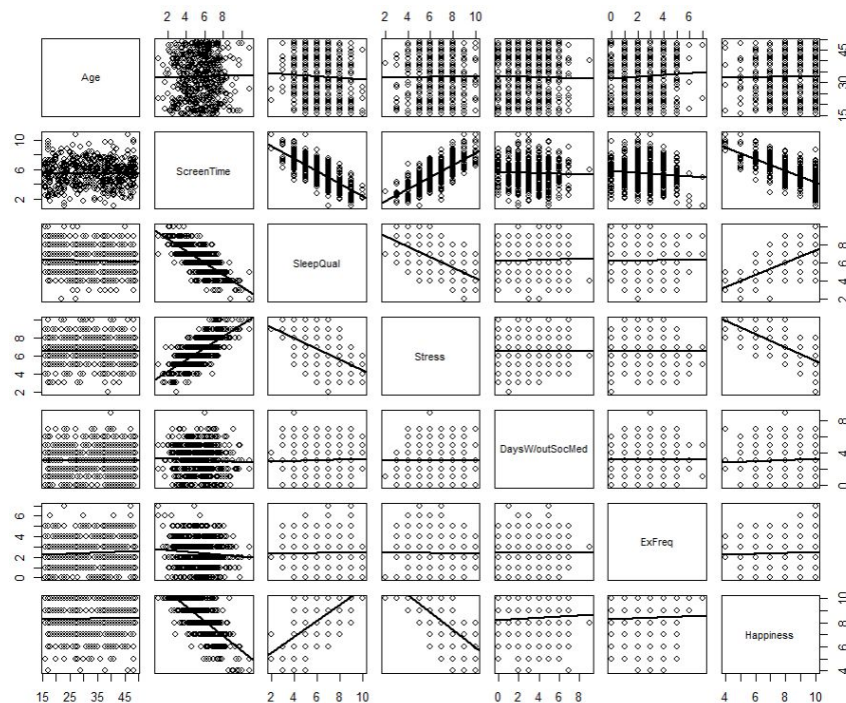
- Principal analysis, Age predicts over 80% of variability
- Clustering (numerical & categorical)
 - Undefined clusters
 - Gower's distance
 - Elbow indicates 3 clusters is best
 - Not very interpretable



Scatterplot matrix and Correlations

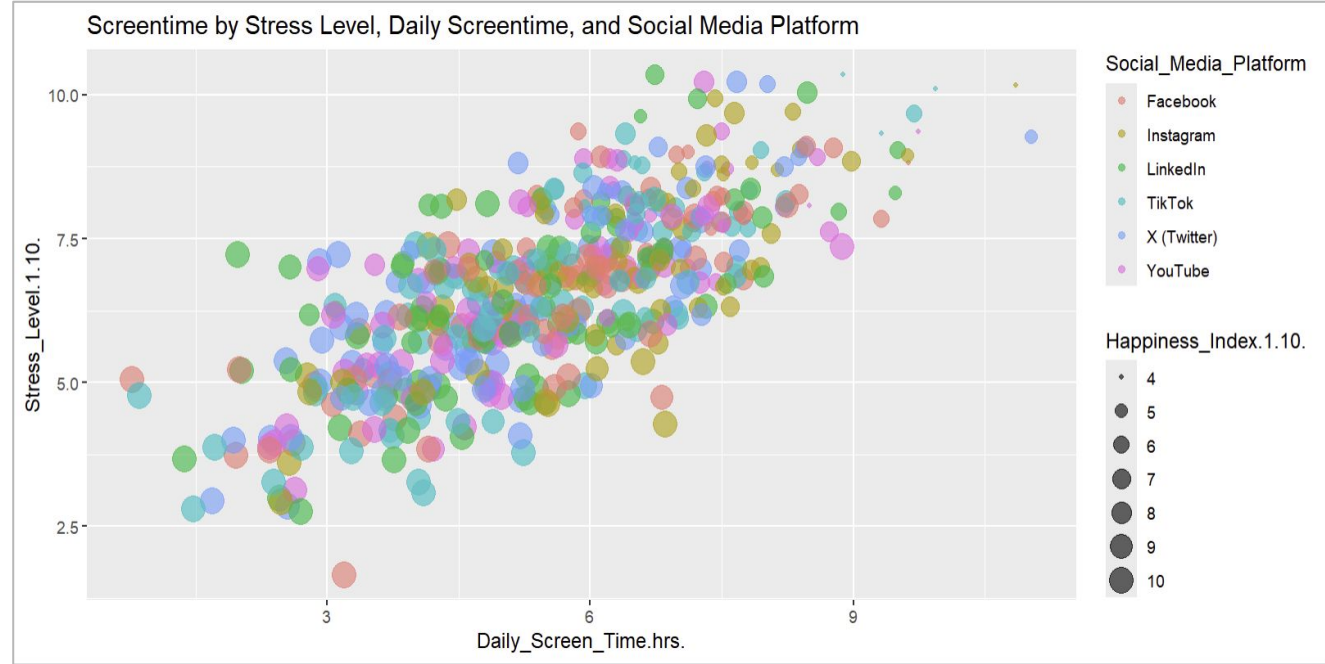
Strongest Correlations between the Variables:

- Screen time / sleep quality : negative (-0.76)
- Screen time / stress : positive (0.74)
- Screen time / happiness : (-0.71)
- Happiness and sleep quality : (.68)
- Happiness and stress: (-.74)
- Stress and sleep quality : (-.58)



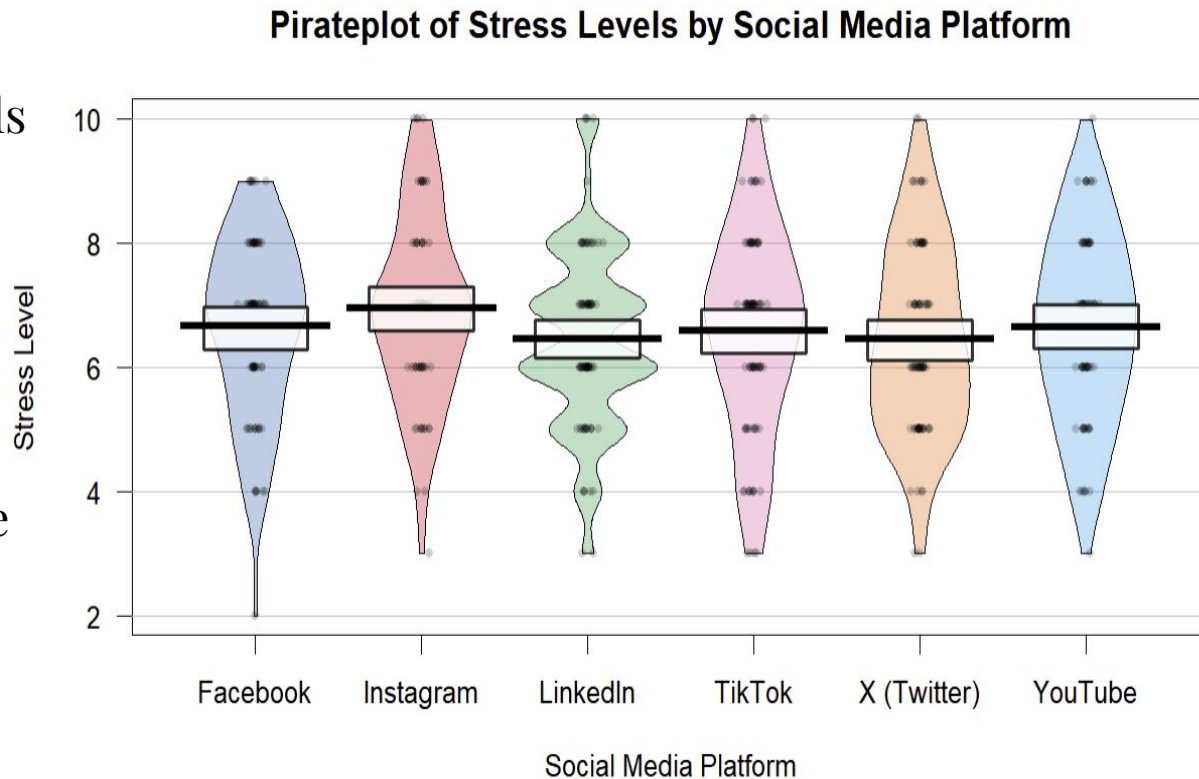
Bubble plot

- Stress level, daily screen time hours, happiness index, and social media platform



Pirateplot

- Showing stress levels by social media platform
- Instagram and Facebook have the highest median stress level
- LinkedIn and X have the lowest median stress level



Factor Analysis Overview

7 numeric variables measured:

- age (years)
- daily screen time (hours)
- sleep quality (1-10)
- stress level (1-10)
- days without social media (per week)
- exercise frequency (per week)
- happiness index (1-10)

Test of the hypothesis that 3 factors are sufficient.

The chi square statistic is 5.68 on 3 degrees of freedom.

The p-value is 0.128

Choice of using 3 factors validated by insignificant p-value (greater than 0.05)

Interpreting Factors

Loadings:

	Factor1	Factor2	Factor3
Age			0.161
ScreenTime	0.771	-0.418	
SleepQual	-0.631	0.674	-0.222
Stress	0.995		
DaysW/outSocMed			
ExFreq		0.134	0.370
Happiness	-0.760	0.290	

Factor 1: Mental health and
emotional wellbeing

Factor 2: Daily habits and lifestyle

Factor 3: General physical health

Factor Analysis Conclusion

Uniquenesses:

	Age	ScreenTime	SleepQual	Stress
DaysW/outSocMed	0.974	0.229	0.098	0.005
0.995				
	ExFreq	Happiness		
	0.844	0.339		

	Factor1	Factor2	Factor3
SS loadings	2.562	0.740	0.215
Proportion Var	0.366	0.106	0.031
Cumulative Var	0.366	0.472	0.503

Explanation of variance in dataset:

- Factor 1: 36.6%
- Factor 2: 10.6%
- Factor 3: 3.1%
- Total: 50.3%

Days without social media and age variables – highest weakness & contribute least to analysis

Conclusion

- Every platform causes a net negative effect
- Increase usage caused a negative effect on mental health
 - Lower sleep quality and stress level